

1 What is regularization in machine learning? why is it needed?

- Regularization is a techniques used to reduce overfitting. by adding a penalty term to the model's loss function.
- Prevents overfitting
- Improves model generalization
- Reduces model complexity
- Handles multicollinearity

2 Difference between Ridge Regression (L2) and Lasso Regression (L1)

Feature	Ridge (L2)	Lasso (L1)
Penalty	Sum of squared coefficients	Sum of absolute coefficients
Formula	$\lambda \sum b^2$	$\lambda \sum b $
Feature selection	No	Yes
Coefficients	Reduced but not zero	Can become zero
Best for	mult. important features	Automatic feature selection

3 What is cross-validation and why is it important?

- Cross-validation is a technique used to evaluate a model's performance on unseen data.
- Detects overfitting.
- Gives reliable performance estimates.
- Better use of available data.
- Helps in model selection.

4 Cross-validation techniques

a) k-Fold cross-validation.

- Dataset divided into k equal parts (folds).
- Train on $k-1$ folds, test on 1 fold.
- Repeat k times and average results.
- e.g. $k=5 \rightarrow 5$ training/testing rounds.

b) Stratified k-fold cross-validation.

- Similar to k -fold.
- maintains the same class distribution in every fold.

used for: classification datasets with m classes.

C) leave - one - out cross - validation (loo (✓))

- special case of k-fold where $k = \text{number of samples}$.
- Train on all samples except one.

Advantage: maximum training data used.

Disadvantage: computationally expensive.

D) Time Series split

- used for time-dependent data.
- ~~But~~ Preserves chronological order.

Train: month 1 → Test: month 2

Train: month 1-2 → Test: month 3

Train: month 1-3 → Test: month 4.

5 Why are Tree-based models less sensitive to feature scaling?

- Tree-based models split data based on feature thresholds, not distances.

Reasons:

- Use comparison rules.
- Feature magnitude does not affect splits.
- No gradient-based optimization. Dependent on scale.